

Exploring Cost Effective Traffic Monitoring Through Kalman Filter Based Fusion of Fiber-Optic Seismology and Computer Vision Tracking Data

Abstract—Fiber optic infrastructure is currently ubiquitous in modern urban environments. Distributed Acoustic Sensing (DAS) presents a viable mechanism for leveraging this pre-existing infrastructure to acquire high-fidelity traffic statistics. To overcome the cost and environmental vulnerabilities of dense camera networks, we propose a method that leverages existing fiber optic infrastructure, potentially supplemented by a minimal number of additional cameras, to achieve single-vehicle tracking for traffic management. Specifically, we implement a Kalman filter framework to fuse DAS-derived acoustic signatures with visual data, enhancing vehicle tracking capabilities. We validated this approach through a custom simulation environment followed by real-world experimental evaluation. Simulation results show that our Fusion approach remains relevant for at least 100 meters beyond the camera’s field of view, and demonstrate that increasing camera density beyond one every 75 meters yields negligible improvements in tracking performance. Furthermore, across 5 randomized trials, the system maintained a mean localization error below 1.5 meters for a distance of 100 meters beyond the camera’s Field of View (FOV). Experimental validation confirmed these findings, maintaining on average the 1.5-meter accuracy threshold for the final 35 meters when we artificially cut the camera view from our video past the 55 meters mark.

Index Terms—IEEEtran, journal, L^AT_EX, paper, template.

I. INTRODUCTION

THE optimization of urban traffic management systems has become a critical priority for modern smart cities, increasingly relying on the continuous, high-fidelity monitoring of vehicle trajectories to mitigate congestion and enhance safety, and many efforts have been devoted to this subject in recent years [9]. While traditional computer vision (CV) systems utilizing roadside cameras have achieved strong local precision in object detection and localization, they remain inherently limited by their “line-of-sight” dependency. Consequently, maintaining continuous tracking across extended traffic corridors necessitates dense hardware deployments, incurring substantial installation, networking, and maintenance costs. Furthermore, optical sensors are susceptible to performance degradation under adverse environmental conditions—such as heavy fog, rain, or low-light scenarios—where gaps in visual coverage inevitably lead to track fragmentation and loss of vehicle identity; it is precisely in these compromised environments that the robust efficacy of sensor fusion is most effectively demonstrated [10].

In parallel, Distributed Acoustic Sensing (DAS) has emerged as a promising alternative, capable of utilizing existing telecommunication fiber optic infrastructure to capture

road vibrations over distances spanning tens of kilometers. This technology effectively turns the fiber into a dense array of sensors, operating with minimal additional hardware and remaining robust against lighting and weather variations [11] [5]. However, despite its wide coverage capabilities, DAS data suffers from significantly lower spatial resolution and higher signal-to-noise ratios compared to visual sensors, often requiring complex signal processing to extract meaningful trajectory data [5].

The primary objective of this work is to design, implement, and evaluate a robust sensor fusion pipeline that effectively integrates DAS-derived detections with CV-based visual tracking. The overarching aim is to maintain tracking continuity across varying camera setups by leveraging the complementary strengths of both modalities [12][9]. Specifically, our goals are to Develop the Stochastic Estimation Architecture, Build a Modular Signal Processing Pipeline and to confirm via Real-World Deployment.

To address these challenges, we develop a comprehensive modular pipeline (15). The system comprises a customized CFOS inference, model for extracting events from raw DAS waterfall plots[5], and the state-of-the-art YOLOv8 algorithm for high-confidence CV-based bounding box detections [15][17]. A critical component of our approach is the manual optical fiber (OF) channel-to-pixel calibration, which maps the one-dimensional fiber coordinates to the two-dimensional camera plane [18]. The independent outputs from YOLO and CFOS are subsequently fed into a centralized Kalman filter pipeline, which synthesizes the final system detections. Additionally, prior to real-world validation, we utilize a simulation environment to rigorously evaluate design trade-offs—such as optimal camera spacing and the maximum viable distance for blind tracking—ensuring the system’s theoretical viability.

II. THEORETICAL BACKGROUND

A. Seismic Waves

Seismic waves include body waves and surface waves (i.e., Rayleigh, Love). In urban traffic contexts, surface waves dominate and decay cylindrically with distance from the source. Vehicle loading excites low-frequency (below a few hertz) quasi-static components, as well as higher-frequency content associated with road roughness.

According to Flamant-Boussinesq Approximation (quasi-static deformation of a subsurface from a point load at a some

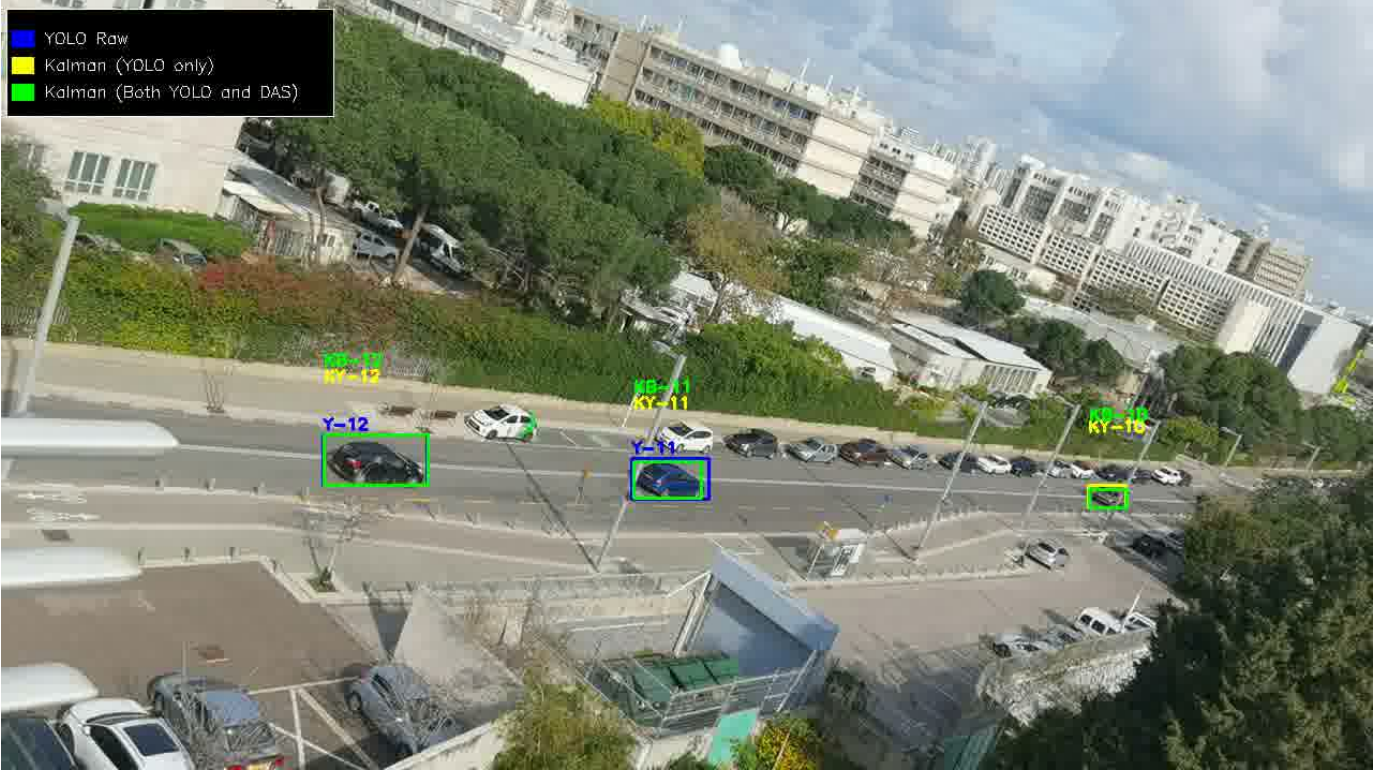


Fig. 1: Demonstration of our Fusion Architecture output projected back to the video captured during the experiment. Kalman filter assists both YOLO and our Fusion outside the camera’s FOV.

location) the displacement u_x on x direction on the ground, as a response to a point load at the origin, at x, y, z , is:

$$u_x(x, y, z) = \frac{F}{4\pi G} \frac{x}{r^2} \left(\frac{z}{r} + \frac{2v+1}{1+\frac{z}{r}} \right) \quad (1)$$

where x, y, z are 3D spatial coordinates, r is the distance between the point load and the origin, F is the force applied on the ground at the point load, G is the shear modulus of the ground, which reflects how sensitive the medium is to deformation, and is assumed to be spatially uniform, v is Poisson’s Ratio of the medium.

B. Fiber-Optic Sensing and Seismology

Rayleigh Scattering’s forward component contributes to attenuation along the fiber, while the backward component underlies techniques such as Optical Time-Domain Reflectometry (OTDR) and Distributed Acoustic Sensing (DAS). OTDR provides a static attenuation profile, serving as a diagnostic tool to observe changes (e.g., slow decline for regular scattering, sharp decline for a damaged line). When external vibrations (such as Seismic Waves) slightly stretch or compress the fiber, the phase of the Rayleigh Scattering backscattered light changes. Unlike OTDR, A DAS interrogator uses narrow-linewidth lasers and advanced phase demodulation to produce a continuous, spatially distributed measurement of the Rayleigh backscattered light’s phase change to sense dynamic strain or strain-rate changes along the fiber.

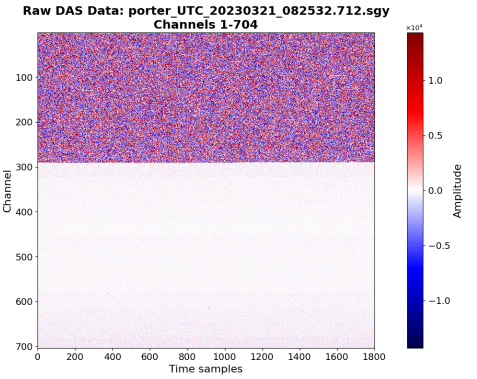


Fig. 2: DAS output example, 21/03/2023 08:25:32-08:26:32, 704 channels, values symmetrically clipped to the ± 99 th percentile. Initial channels are noisy

C. DAS Measurement

The DAS system measures the average change in strain (or strain rate) along the Gauge Length, denoted as L . Assuming the seismic waves are constant on a single Gauge Length, a DAS recording of strain change can be modeled as a spatial derivative of the subsurface quasi-static deformation u_x averaged along L :

$$\varepsilon_{DAS} = \frac{1}{L} \left[u_x \left(x + \frac{L}{2}, y, z \right) - u_x \left(x - \frac{L}{2}, y, z \right) \right] \quad (2)$$

In our experiment the DAS was configured to measure the strain rate change, this can be modeled as a spatial derivative

of the strain change measurement model:

$$\dot{\epsilon}_{\text{DAS}} = \frac{1}{L} \left[\dot{u}_x(x + L/2, y, z, t) - \dot{u}_x(x - L/2, y, z, t) \right].$$

The output of a DAS measurement 2 is a 2D array of the strain-rate change at all the recorded channels (Gauge Length segments) of the optic fiber in time. We refer to a Gauge Length segment of the Optic Fiber as a Channel[20][21][22].

D. Computer Vision (Camera)

The visual subsystem utilizes a stationary surveillance camera to provide high-precision localization of vehicles in the near-field. The processing pipeline consists of object detection followed by a projective transformation to map image coordinates to the road frame. We employ the YOLO architecture, specifically the YOLOv8 model to detect 2 Classes of vehicles (Cars and Busses) in the video feed[23][26].



Fig. 3: A frame from the video taken during the experiment, 08:26:04 21/03/2023.

III. METHODS

A. YOLO Detection Pipeline

The YOLOv8 model outputs a bounding box with a confidence score $c_{det} \in [0, 1]$, using a custom threshold of 0.5 (4). We utilize the geometric center (u_{center}, v_{center}) as the representative tracking point[24].

Note: Using the box center aligns with the visual centroid but introduces a systematic parallax offset during planar projection, unlike the more common bottom-center road contact point.

The tracking pipeline filters out stationary objects by enforcing a minimum, sustained 2D pixel displacement over the object's tracked life cycle. This is a post-processing step applied after a track has terminated.

For each completed track, the process calculates the total displacement by comparing a track's estimated start position to its estimated end position. Both the start and end positions are computed by averaging the coordinates of a small, configurable window of detections to mitigate noise and jitter. Further details in the appendix under YOLO Stationary Object Removal Algorithm.

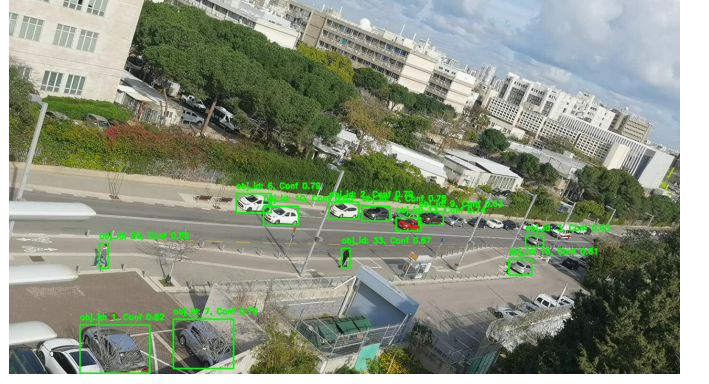


Fig. 4: A frame from the YOLO algorithm output, multiple Bounding Boxes can be seen on moving vehicles and on stationary vehicles.

Coordinate Projection (Homography): To fuse visual data with DAS measurements, the 2D pixel coordinates (u, v) must be transformed into the physical road coordinate system (x, y) . Assuming the road surface is locally planar, we utilize a perspective transformation matrix (Homography), denoted as $\mathbf{M} \in R^{3 \times 3}$. The mapping from the image plane to the world frame is given by:

$$\begin{bmatrix} wx \\ wy \\ w \end{bmatrix} = \mathbf{M} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \quad (3)$$

where w is a scaling factor. This matrix $\mathbf{M}$ is calibrated offline using known ground control points on the road surface.

Error Model: Geometric and Confidence-Weighted Noise: The measurement uncertainty for the visual sensor is modeled as a dynamic function of two factors: geometric degradation and detection confidence.

First, due to perspective projection, quantization error scales non-linearly with longitudinal distance x , reducing precision at the limit of the FOV. Second, the YOLO confidence score c_{det} serves as a proxy for detection quality.

We construct the measurement noise covariance $\mathbf{R}_{cam}$ by combining these factors, such that the uncertainty σ_{cam}^2 increases with distance and decreases with higher detection confidence[25]:

$$\sigma_{cam}^2(x, c_{det}) \propto \frac{f(x^2)}{c_{det}} \quad (4)$$

This ensures that the Kalman Filter relies less on visual updates when the vehicle is far away or when the detection probability is low, preventing low-quality visual data from corrupting the state estimate.

1) Validity of the Gaussian Assumption: Strictly speaking, object detections from a neural network like YOLO do not adhere to the linear-Gaussian requirements of the Kalman Filter. The measurement error is dominated by *quantization* (discrete pixel coordinates), which follows a uniform distribution rather than a normal distribution, and *nonlinearities* such as bounding box jitter and occasional false positives (heavy-tailed outliers).

However, we approximate this error profile as Gaussian because the aggregation of multiple independent error sources

(sensor noise, quantization, perspective distortion) tends to converge towards a normal distribution via the Central Limit Theorem.

B. CFOS Detection Pipeline

The DAS raw system output, a 2D array (discrete time vs. optic fiber channels), was first pre-processed according to the requirements of the pre-trained CFOS Neural Network (NN) according to the flow described in 14.

The raw DAS output (.sgy files) are inputted in batches of 12 files/minutes to the preprocessing pipeline. First the spatial and temporal medians are removed, then the data goes through a Low Pass Filter, afterwards values are clipped to the positive and negative 99th percentile, followed by downsampling from 1KHz to 30 Hz. The information on the DAS measurement should remain intact since we know from the geo-physical theoretical background that the quasi static components excited by vehicle point loading is in bellow 10 Hz frequencies (The downsampled frequency is higher than the Nyquist sampling frequency of the phenomenon - as a safety margin). Next channels 280-540 were selected, and an F-K filter was applied. Lastly we removed the first and the last DAS files (minute each), and output 10 files/minutes of preprocessed DAS data. 5 shows an example of the result of this signal processing flow.

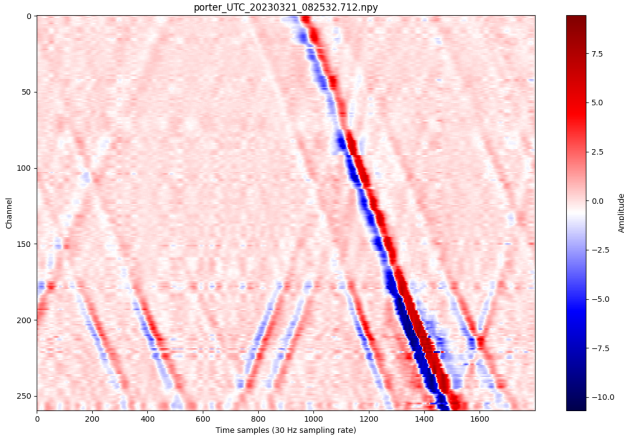


Fig. 5: DAS preprocess output, 08:25:32-08:26:32, 21/03/2023, channels 280-370

The CFOS NN allows a range of output channels to be picked, and the output dimensions are (number of tracked channels)x(time). For each point (*channel*,*time*) the CFOS outputs a probability for each class that there is a class object in this coordinate. Specifically, we used a checkpoint that was trained to identify two classes - Cars and Busses. These probabilities are processed by a confidence lower bound, followed by a contour identification with a minimal area to reduce noise. From this output we extract vehicle positions with a known noise profile 6.

The Adaptive Noise Covariance (ANC) matrix, $\Sigma_n(t)$, CFOS output is modeled as linearly dependent on the inverse

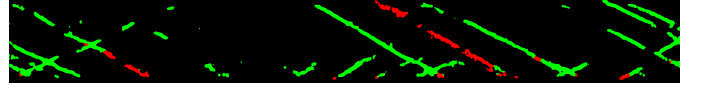


Fig. 6: CFOS output visualized, Red for Busses, Green for Cars, channels 280-480

of the contour channel-width $\Delta w(t)$ at the time of detection, scaled by the detection probability $P_d(t)$ at that channel, and summed with a constant offset term based on the known properties of the model error. This relationship is formalized by the following equation:

$$\Sigma_n(t) = C_1 \left(\frac{1}{\Delta w(t)} \cdot P_d(t) \right) + C_0 \mathbf{I} \quad (5)$$

where: $\Sigma_n(t)$ is the Adaptive Noise Covariance matrix at time t . $\Delta w(t)$ is the contour channel-width at time t . $P_d(t)$ is the detection probability at the detection channel at time t . C_1 and C_0 are empirically determined scaling coefficients. $\mathbf{I}$ is the identity matrix, which ensures a minimum base covariance.

C. Sensor Fusion via Kalman Filter

1) *Fundamentals*: The Discrete Kalman Filter (DKF) is a recursive estimator that provides an optimal estimate of a system's state by minimizing the mean squared error for linear systems corrupted by Gaussian noise. It is widely used for state estimation in dynamic systems, such as tracking, navigation, and control [3], [2]. The fundamental operation of the filter is a two-step cycle: Time Update (Prediction) and Measurement Update (Correction). Further details in the appendix under DKF Fundamentals.

2) *Appliance*: To apply this framework to vehicle tracking, we approximate the vehicle motion and sensor errors as linear systems with Gaussian characteristics. We define a 2D state vector $\mathbf{x}$ comprising position (p), velocity (v), and acceleration (a) in longitudinal (x) and lateral (y) directions:

$$\mathbf{x} = [p_x, p_y, v_x, v_y, a_x, a_y]^T \quad (6)$$

A critical challenge is the intermittent availability of sensor data (e.g., occlusion or blind zones). To address this, we employ a dynamic update strategy. At each timestep k , we inspect the available valid detections and construct the measurement vector $\mathbf{z}_k$, the observation matrix $\mathbf{H}_k$, and the noise covariance $\mathbf{R}_k$ *ad hoc*: Simultaneous Data (DAS + Camera): When both sensors provide valid detections, we stack the measurements into a composite vector $\mathbf{z}_k = [\mathbf{z}_{cam}^T, \mathbf{z}_{das}^T]^T$. The observation matrix $\mathbf{H}_k$ is similarly concatenated, and $\mathbf{R}_k$ is constructed using the specific uncertainty (σ^2) of each sensor type.

Single Source (DAS or Camera only): If only one modality is available, $\mathbf{z}_k$ reduces to the dimension of that single measurement, and $\mathbf{H}_k$ adapts to map the state exclusively to the active domain.

No Data (Prediction Only): If no detections are available (e.g., total occlusion), the measurement update is skipped entirely. The system relies solely on the prediction step ($\hat{\mathbf{x}}_{k|k-1}$) to propagate the state until new data arrives.

IV. EXPERIMENTS

To validate the proposed sensor fusion framework, we conducted a comprehensive analysis based on theoretical simulation and physical experimentation. Our approach utilized a "digital twin" methodology: first, a high-fidelity computer simulator was employed to verify the theoretical performance of the Kalman filter under controlled noise conditions. Crucially, this simulation was configured to replicate the geometry and environmental conditions of our real-world test site. Subsequently, the method was evaluated using data collected from a Distributed Acoustic Sensing (DAS) system with a surveillance camera in the modeled urban environment.

A. Simulation Environment

A simulation environment was developed to isolate the impact of specific noise characteristics, to quantify and evaluate the benefits of sensor fusion in scenarios that are difficult to replicate consistently in the field. The simulation modeled a vehicle traveling along the x -axis (longitudinal road direction), in the x -axis direction or against it, depends on the driving lane. The simulation received a initial speed as user input, and changed the vehicle's acceleration randomly each two seconds under restrains of vehicles typical acceleration in urban environment (max 1 m/s acceleration for private, 0.7 m/s for buses and 0.5 m/s for trucks).

To ensure the validity of our comparative analysis, we reconstructed the specific real-world scene where the Real World Experiment was held within the simulator. By running this digital replica, we generated synthetic computer vision results that mirrored the format and behavior of the physical YOLO detections. This allowed us to benchmark the vision pipeline against absolute ground truth before applying it to the real-world data.

1) *Sensor Noise Modeling*: A core assumption of our approach is that the measurement noise for both the camera and DAS sensors can be approximated as Gaussian. This assumption is critical, as the Kalman filter is theoretically proven to be the optimal estimator for linear systems subject to additive white Gaussian noise (AWGN) [1].

2) *DAS Simulation*: The DAS signal was generated by superimposing additive Gaussian noise onto the ground truth vehicle trajectory. The noise variance (σ_{DAS}^2) was calibrated to match the error profile of the specific Convolutional Neural Network (CFOS) used in our real-world experiments [5].

$$\mathbf{x}_{DAS}(t) = \mathbf{x}_{GT}(t) + \boldsymbol{\epsilon}(t), \boldsymbol{\epsilon}(t) \sim \mathcal{N}(\mathbf{0}, \sigma_{DAS}^2 \mathbf{I}).$$

3) *Camera Simulation*: Camera Simulation: Camera error is decomposed into *Spatial Error* and *Temporal Error*.

Spatial Error (Projection + Quantization): Ground-truth 3D points are projected onto the image plane and quantized to pixel centers, then reprojected back to world coordinates. This induces spatially variant error that grows with range, mimicking reduced precision at distance. Further details in the appendix under Camera Simulation Spatial Error Model.

Temporal Error (Lag-only Jitter): Camera timestamps are delayed by half-normal Gaussian jitter and constrained to be monotone with a minimum inter-frame gap. Excess lag is rescaled to preserve the nominal time span, while strictly preventing early frames. This is done to model fluctuations in framerate caused by the camera's internal oscillator frequency deviations. Further details in the appendix under Camera Simulation Temporal Error Model.

YOLO Confidence Simulation and Sigma conversion: The camera uses a distance-based confidence model that simulates YOLO detection confidence. After downsampling vehicles to the camera's frame rate and applying spatial errors, the camera calculates the Euclidean distance (radius) from each detected vehicle's true ground-truth location to the camera position. These radii are then passed through a gaussian_plateau_noise confidence model, which produces values between 0-100. This model maintains near-maximum confidence (≈ 100) for vehicles within a plateau radius R_0 , then drops off with a Gaussian decay controlled by parameters r_{keep} (distance threshold) and C_{th} (confidence threshold at that distance). Gaussian noise ($\sigma_{noise} = 3$) is added for realism. Finally, the confidence value is inverted and scaled ($sig_mult \cdot (1/confidence) + sig_bias$) to produce a sigma (uncertainty) value that's stored in the last column of the vehicle tensor—higher confidence yields lower sigma for the Kalman filter. further detail in appendix under YOLO Confidence Simulation and Sigma Calculation.

4) *Simulation Study A: Extended Range and Sensor Transitions*: To evaluate the system's performance in "blind zones"—areas covered by DAS but outside the camera's Field of View (FOV)—we simulated a 200-meter straight road segment. The camera coverage was limited to the first 90 meters ($x \in [0, 90]$), leaving the subsequent 110 meters ($x \in (90, 200]$) monitored exclusively by the fiber-optic sensor.

This experiment investigates the system's ability to maintain high-precision tracking during sensor transitions. We postulate that if the Kalman filter converges on an accurate state estimate (position, velocity, and acceleration) during the camera-visible phase, it will effectively "carry over" this precision into the DAS-only zone. Consequently, the fused system should maintain superior tracking accuracy even after the camera signal is lost, compared to a system relying on DAS alone. We assessed this by calculating the Mean Absolute Error (MAE) of the filter's predictions versus the ground truth at varying distances from the camera's FOV limit. his analysis allows us to determine the effective range at which the fused system degrades to the performance level of raw DAS-based tracking, thereby identifying the maximum viable distance between camera nodes.

5) *Simulation Study B: Camera Density Optimization*: To determine the optimal balance between tracking accuracy and infrastructure cost, we conducted a sensitivity analysis regarding camera density. We performed iterative simulations with an increasing number of cameras ($N = 1$ to $N = 6$) distributed evenly along the track Figure 7a. The objective was to identify the point of diminishing returns—specifically, the

camera-to-meter ratio at which computer vision metrics (such as HOTA and DetRe and DetA) strictly saturate, yielding no significant performance gain from additional hardware. This informs the minimal infrastructure requirements for scalable deployment.

6) *Simulator Limitations and Modeling Assumptions*: The method with which a simulator camera samples the vehicles in its line of sight is to take all the vehicle locations, compute what points are in the "light pyramid" (similar to light cone, but with rectangular dimensions of the camera output) determined by the modeled display resolution the placement (Roll, Pitch and Yaw), and deleting the rest. since we do not accurately model lighting (ray tracing) it was problematic to model occlusion that would be consistent across cameras viewing the same occlusion from different angles. The simulator assumes the channels are spaced 1 meter from each other - which is correct on the optic-fiber, but practically is incorrect in real world measurements. optic fibers are prone to coiling, looping of leftover fiber at the end of a segment, etc. The DAS based measurement and error in the simulator was developed to model the specific characteristics of the CFOS architecture results [5].

B. Real-World Experiment

Following theoretical validation via the digital twin simulation, we evaluated the system in the corresponding physical urban environment. The experimental setup consisted of two primary sensors: Camera System: A stationary camera monitored a straight road segment (covering $x = 0$ to $x = 150$ meters, where detections degrade significantly beyond $x \approx 90$ meter) filming at 30 frames per second (fps) with a resolution of 1280×720 pixels. DAS System: A DAS system, measuring strain-rate change (not strain change), with a Gauge length $L = 1$ m and a measurement frequency of 1 kHz (temporal sampling rate), connected to an Optic Fiber placed in the ground close to the surface between the edge of the eastern sidewalk to the road on Klauzner Street.

Seismic waves were manually induced using a hammer, and the resulting signals were simultaneously captured by the camera (pixels) and the DAS data. This capture data was then manually processed to establish precise (DAS channel, video pixel) matching points. This spatial calibration, denoted as `channel2pixel_db`, allowed for the accurate conversion from DAS 1D data (channel space) to 2D space (the pixel plane initially following by conversion via a Homography to the road plane - assuming the road is a plane) using interpolation and extrapolation of channel's pixel values.

Fiber-Optic based tracking data was processed using the pre-processing pipeline 14 followed by the CFOS. Simultaneously, the video from the camera feed was processed using the YOLO algorithm to generate Bounding Boxes (BB) on each video's frames as seen in 4. The center pixel of these BB was extracted with the final output being an array of (*frame, centerpixel*) YOLO detections.

V. RESULTS AND DISCUSSION

A. Comparative Baselines

To rigorously assess the contribution of the proposed fusion model, we compared three distinct tracking configurations:

- Fusion Model (Ours): A Kalman filter integrating measurements from both DAS and YOLO.
- YOLO + Kalman Filter: A Kalman filter fed exclusively by camera detections.
- Raw YOLO: The raw, unfiltered output from the object detection algorithm.

This comparative structure was selected to isolate specific performance factors. The comparison against Raw YOLO establishes a baseline against the standard commercial approach. The comparison against YOLO + Kalman Filter is particularly crucial; it ensures a "fair fight" by isolating the specific contribution of the DAS data. Since previous studies have shown that standalone DAS detections are generally less precise than vision-based methods, our goal is to demonstrate that the fusion of DAS - even if individually noisier - improves the robustness and continuity of the vision-based state estimation, particularly in cases of lack of camera nodes density, occlusions or visual noise (such as change of lighting, fog, rain, etc).

B. Simulation Results

1) *Study A: Tracking Continuity in Blind Zones*: To evaluate the system's robustness during sensor transitions, we analyzed the tracking error as the vehicle exits the camera's Field of View (FOV) and enters the DAS-only zone.

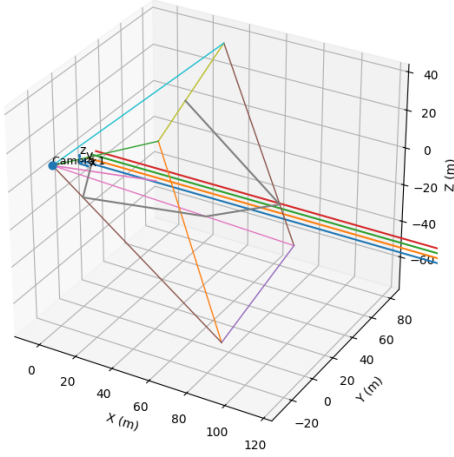
Metric Selection (MAE): While standard computer vision metrics (e.g., MOTA, HOTA) are common for detection, we utilized Mean Absolute Error (MAE) of the estimated position for this analysis. This choice was driven by consistency with our real-world field tests, where Ground Truth (GT) labels were unavailable for the non-visual zones. By utilizing MAE in the simulation—where absolute ground truth is available—we establish a direct performance baseline that mirrors the physical constraints of the field experiment.

Performance Analysis: Figure 8 illustrates the MAE of the Fusion model (Blue) versus Raw DAS detections (Orange) as a function of distance from the camera boundary. The origin ($x = 0$) represents the end of the camera's FOV; negative values ($x < 0$) denote the region covered by both sensors, while positive values ($x > 0$) denote the blind zone monitored only by DAS.

Inside the FOV ($x < 0$): The Fusion model benefits from the high spatial precision of the camera, achieving a negligible error of approximately 0.1m–0.4m. In contrast, the Raw DAS signal in this region exhibits a significantly higher baseline error (≈ 2.4 m), verifying the noisier nature of the acoustic sensor.

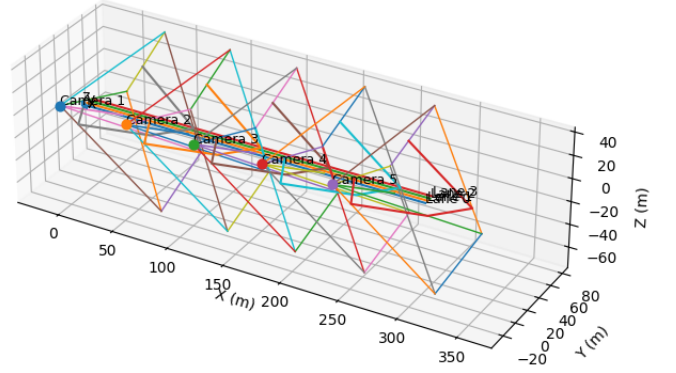
The Transition ($x = 0$): As the vehicle nears the transition point in the simulator, the simulated distance based confidence kicks in and increases the uncertainty that it Inputted to the Kalman Filter, causing a premature rise in MAE. Once a vehicle crosses the $x = 0$ boundary, the Fusion model

Road lanes, camera placement, and vision frustums



(a)

Road lanes, camera placement, and vision frustums



(b)

Fig. 7: Simulator.

(a) 1 camera sep-up, a virtual twin of our experiment model. (b) 5 cameras sep-up. FIXME

experiences an expected increase in error variance, jumping from $\approx 0.4\text{m}$ to $\approx 1.1\text{m}$ MAE. However, crucially, it does *not* spike to the level of the Raw DAS immediately.

Blind Zone Persistence ($x > 0$): throughout the entire 100m simulated extension, the Fusion model maintains a stable error profile, fluctuating between 1.0m and 1.5m. This is significantly lower than the Raw DAS error, which oscillates around 2.0m – 2.8m.

Conclusion on Effective Range: The results confirm the hypothesis that the Kalman filter effectively “carries over” precision from the visual domain. Even 100 meters past the camera’s limit, the fused state estimate’s MAE holds roughly 50% lower than the raw acoustic measurements (MAE $\approx 1.2\text{m}$ vs $\approx 2.4\text{m}$). Part of it can be attributed to the addition of the Kalman filter to the raw DAS data. But it also implies that in a sparse infrastructure setup, camera nodes can be spaced with significant gaps (at least 100m) while the system maintains a tracking precision significantly superior to standalone DAS.

2) Simulation Study B: Infrastructure Optimization and Sensitivity Analysis: To determine the minimal infrastructure required for reliable tracking, we conducted a sensitivity analysis by varying the number of camera nodes (N) uniformly distributed along the 300m simulated corridor. We compared the Fusion Model against the Vision-Only Baseline (YOLO + Kalman Filter) and Raw YOLO.

The results can be described by three distinct performance regimes.

a) The Sparse Regime ($N = 1$): The Necessity of Fusion: In low-density setups, where camera coverage is intermittent, the Fusion framework demonstrates dominant superiority.

Quantifiable Gain: With a single camera ($N = 1$), the Fusion model achieves a HOTA score of 85.45, whereas the Vision-Only baseline reaches only 37.77 (Figure 9). This represents a $> 100\%$ performance improvement. **Mechanism of Improvement:** The disparity is driven primarily by

Detection Recall (DetRe). As shown in Figure 10, the Fusion model maintains a recall of 90.36% even with minimal visual coverage, while the vision-only system drops to 37.24%. This confirms that in sparse setups, the DAS subsystem effectively “bridges the gaps.”

b) The Saturation Point ($N = 4$): The “Knee” of the Curve: As infrastructure density increases to $N = 4$, we observe the point of diminishing returns. The Vision-Only performance improves dramatically, with HOTA rising to 85.92, virtually matching the Fusion score of 88.17 (Figure 9). This saturation is explained by the Recall metric (Figure 10): at $N = 4$, the visual coverage reaches 88.15%, nearly equaling the Fusion recall of 93.08%. Consequently, the DAS subsystem provides no additional spatial information, rendering it redundant.

c) The Dense Regime ($N = 6$): The Cost of Noise: A critical and counter-intuitive phenomenon occurs at high densities ($N = 6$). While the Vision-Only model maintains high performance (HOTA 83.20), the Fusion model’s performance significantly degrades to HOTA 54.95 (Figure 9).

Explanation: This degradation is due to Noise Injection. When the road is fully covered by high-precision cameras ($N = 6$), the Kalman Filter receives near-perfect visual measurements at every timestep. Incorporating DAS data—which is inherently noisier and less spatially precise—into the update step introduces unnecessary variance to the state estimate.

Implication: This result validates our core hypothesis: DAS is a *complementary* sensor for continuity, not a *replacement* for precision.

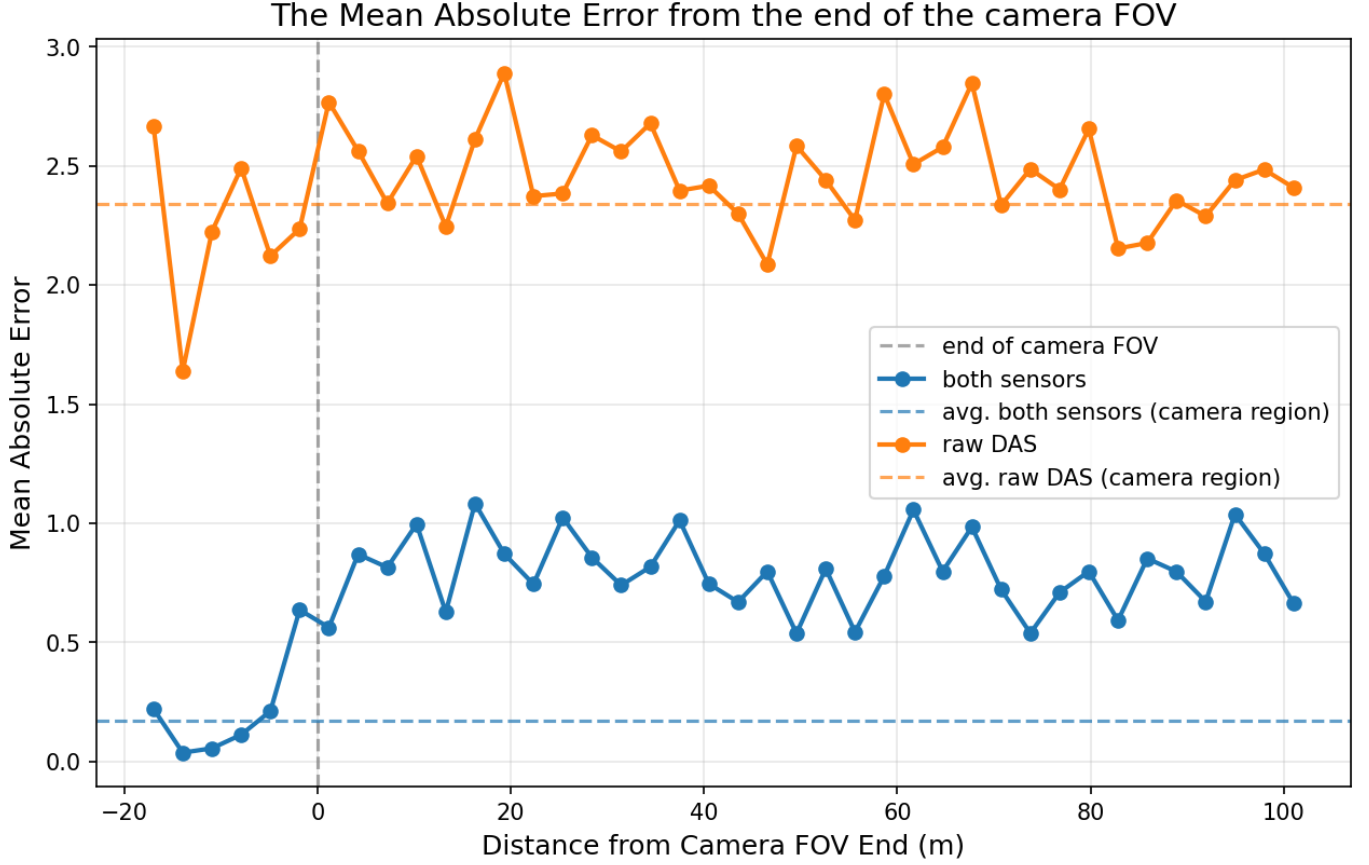


Fig. 8: Study A (Blind Zone): The Fusion model (Blue) maintains significantly lower error than raw DAS (Orange) even 100m after the camera FOV ends at $x=0$.

C. Real-World Experimental Results

Following the simulation validation, we evaluated the system on the physical testbed. This section presents the quantitative performance of the Fusion framework compared to the baselines.

1) *Quantitative Analysis: Tracking Metrics*: To rigorously assess the contribution of the DAS sensor, we compared our Fusion Model against two baselines: Raw YOLO (standard commercial detection) and Vision-Only Kalman Filter (YOLO + KF). The inclusion of the Vision-Only KF is critical; it ensures a “fair fight” by granting the camera baseline the same motion-smoothing advantages as the fusion model, thereby isolating the specific value of the DAS data.

a) *Metric Comparison*:: Table I and Figure 11 summarize the performance. Even though static objects were manually removed to optimize the Raw YOLO baseline, the Fusion model achieved the highest performance across all key metrics.

Overall Accuracy (HOTA): The Fusion model achieved a HOTA score of 55.25, outperforming the Vision-Only KF (54.53) and significantly surpassing Raw YOLO (49.09). While the margin between Fusion and Vision-Only is tighter than in simulation (due to the dense camera coverage in the test zone), the improvement confirms that DAS positively

contributes to the state estimate.

The Impact on Recall (DetRe): The primary mechanism of improvement is visible in the Detection Recall score. The Fusion model reached 57.55% recall, compared to 55.43% for Vision-Only and 43.58% for Raw YOLO. This increase demonstrates the Gap-Filling Capability of the system: even within the camera’s general FOV, momentary occlusions or missed detections by YOLO are compensated for by the continuous DAS signal, preventing track fragmentation.

2) *Quantitative Analysis: Blind Zone Continuity (MAE)*: A key objective of this work is to determine if the Kalman filter can maintain a “good” prediction when the camera signal is lost, relying on DAS to prevent drift. To measure this in the real world—where ground truth is unavailable outside the camera’s view—we devised an Artificial Blind Zone Experiment. We artificially discarded all YOLO detections beyond $x = 55\text{m}$, treating the region $x \in [55, 90]$ as a DAS-Only zone. This allowed us to compare the tracker’s performance against the available Ground Truth in that final 35m segment.

Figure 12 illustrates the Mean Absolute Error (MAE) of the Fusion prediction versus the Raw DAS measurement relative to the cut-off point.

Results: Suppression of Noise: The Raw DAS signal (Purple) exhibits high variance, with error spikes reaching

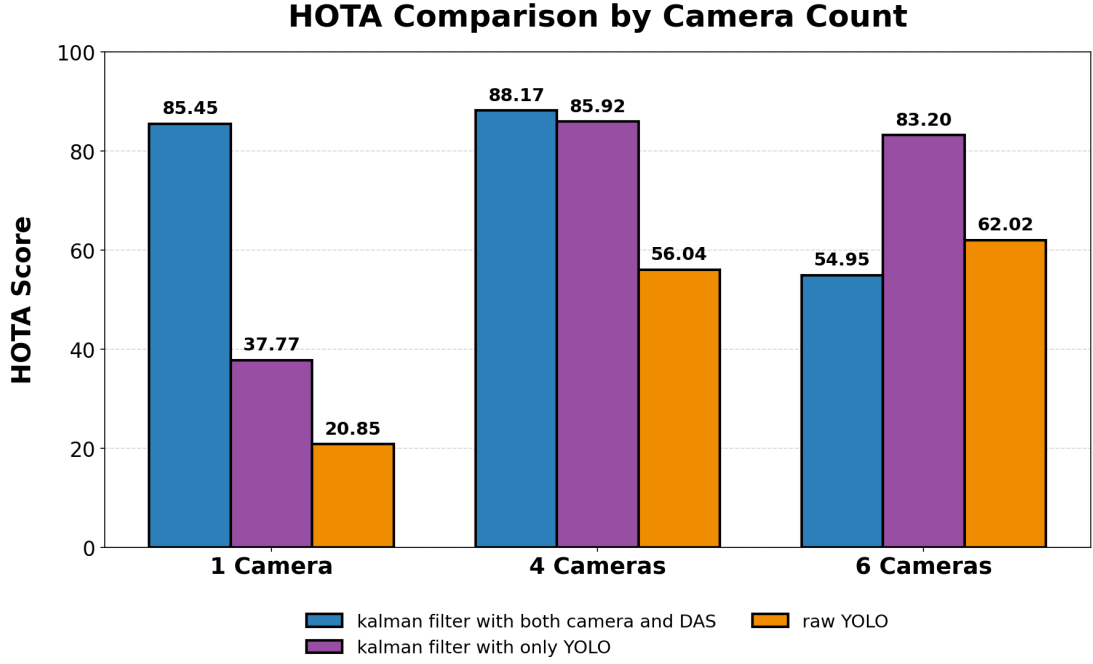


Fig. 9: HOTA results at different camera density: HOTA at $N = 1$: Fusion Model (85.45) dominates Vision-Only (37.77), demonstrating the necessity of fusion in sparse networks. HOTA at $N = 4$: Performance saturates. Fusion (88.17) is only marginally better than Vision-Only (85.92). HOTA at $N = 6$: high camera density, the Vision-Only model (83.20) exceeds the Fusion model (54.95). This degradation is attributed to noise injection from the lower-precision DAS sensor.

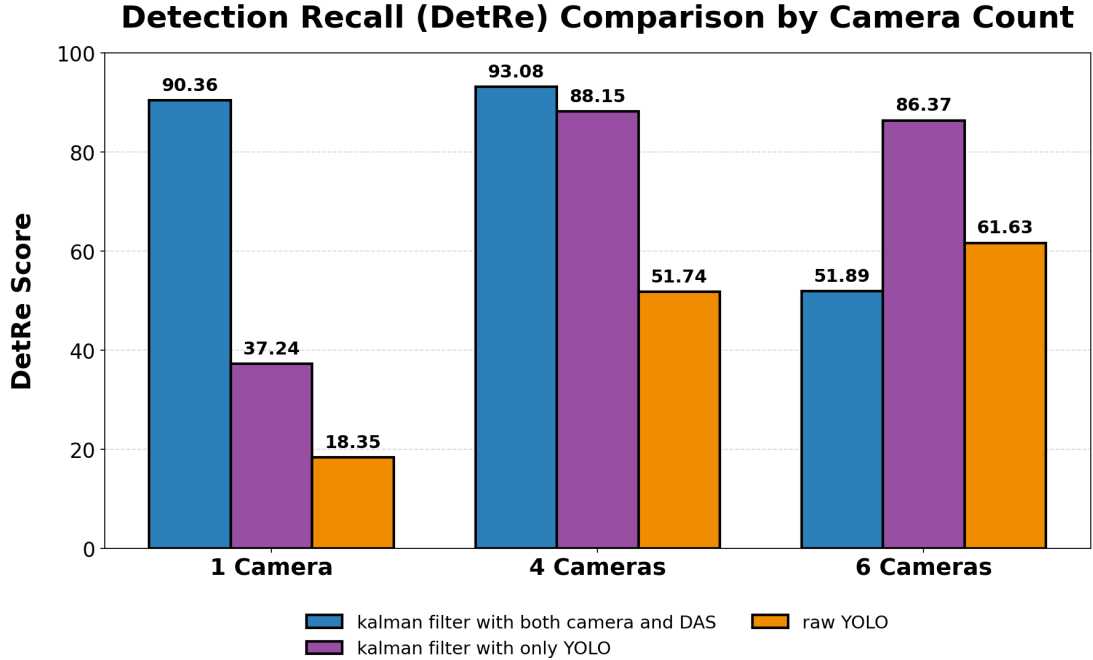


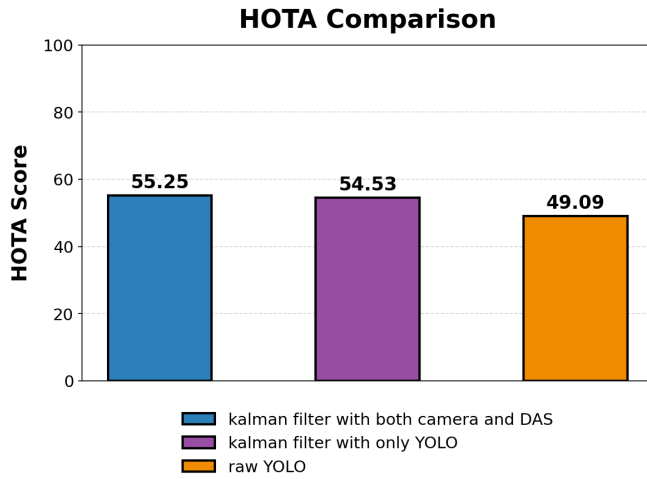
Fig. 10: DetRe at $N = 1$: The Fusion Model maintains high recall (90.36%) by using DAS to bridge the single camera's blind spots. DetRe at $N = 4$: Visual coverage reaches 88.15%, nearly equaling Fusion recall (93.08%). DetRe at $N = 6$: Not only diminishing returns, but a substantial decrease in Fusion HOTA score.

2.5m–2.8m. In contrast, the Fusion model (Orange) maintains a significantly smoother trajectory with lower error (avg ≈ 1.2 m).

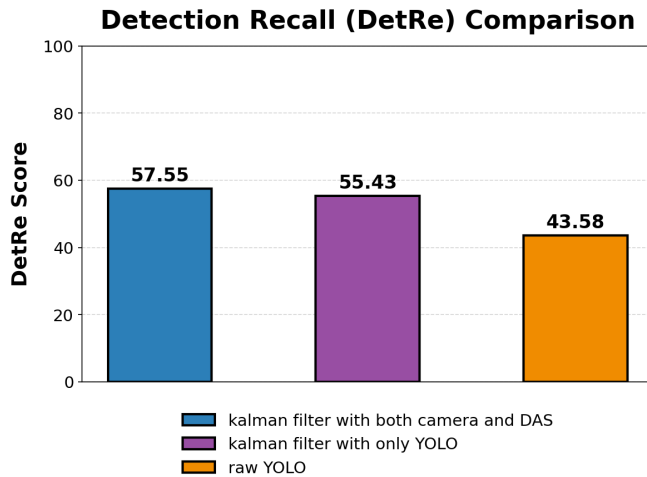
Persistence of Precision: As the vehicle enters the artificial blind zone ($x > 55$), the Fusion error does not immediately

TABLE I: Real-World Experiment Performance Metrics Results of our Comparison Baselines

Tracker	HOTA $\uparrow$	MOTA $\uparrow$	DetRe $\uparrow$	TP	FN	FP
Fusion (Ours)	55.25	58.99	57.55	2539	960	475
Vision-Only (KF)	54.53	60.16	55.43	2483	1016	378
Raw YOLO	49.09	54.42	43.58	1995	1504	91



(a) HOTA Comparison: Fusion yields the robustest tracking.



(b) Recall Comparison: DAS improves detection count (TP).

Fig. 11: Real-World Performance.

The Fusion model (Blue) outperforms the optimized Vision-Only baseline (Purple) and Raw YOLO (Orange), primarily by improving Recall (DetRe) through DAS inputs.

jump to the DAS noise level. Instead, the filter effectively utilizes the velocity state converged during the visual phase ($x < 55$) and updates it with the noisy but trending DAS measurements. This confirms that the Fusion framework acts as a low-pass filter for the DAS signal, offering a viable solution for extending tracking range beyond the camera’s physical limit.

3) *Qualitative Analysis*: To provide context to the statistical metrics, we analyzed specific real-world scenarios where visual tracking typically fails. We visualized the system’s output on video overlays and geospatial maps to verify robustness.

Scenario 1: Occlusion and Visual Clutter

We examined a video sequence of the test site where the vehicle passes through areas of high visual clutter (e.g., parked cars, trees, and static infrastructure). Figure 13a illustrates a representative frame comparing three tracking outputs: Raw YOLO (Blue), Vision-Only Kalman Filter (Yellow), and Fusion Model (Green).

Baseline Stability (Vision-Only vs. Raw): The video analysis confirmed that the Vision-Only KF (Yellow) significantly outperforms Raw YOLO (Blue). In instances where Raw YOLO incorrectly switched IDs between the moving vehicle and a static background object, the Vision-Only filter utilized its Constant Acceleration (CA) motion context to reject the outlier and maintain the track.

Fusion Superiority (Fusion vs. Vision-Only): The Fusion Model (Green) demonstrated decisive robustness in two critical failure modes:

- 1) The “Distraction” Case: When the visual detector “locked” onto a static background object and ceased updating the vehicle’s position, the Vision-Only filter eventually stalled or drifted. In contrast, the Fusion model utilized the continuous DAS measurements to “pull” the state estimate along the true trajectory, effectively ignoring the visual false positive.
- 2) The “Total Loss” Case: In segments where visual detection failed entirely (False Negatives), the Vision-Only filter relied solely on prediction (dead reckoning), leading to drift. The Fusion model, however, continued to receive valid (albeit noisier) updates from the DAS subsystem, maintaining a valid active track where the camera-only system saw “nothing.”

Scenario 2: Extended Range and Continuity (Klauzner Street)

We further validated the tracking continuity on the specific topology of our test site (Klauzner Street, Tel Aviv). Figure 13b shows the projected Kalman Filter predictions (Green dots) overlaid on a Google Maps satellite view.

The camera coverage effectively ends at $x = 90$ m. Beyond this point, the system enters the blind zone. The geospatial plot reveals that the Fusion framework transitions smoothly from the visual-dominant phase to the DAS-dominant phase. The trajectory points in the blind zone do not scatter chaotically (as raw DAS detections often do) but form a coherent path that aligns with the road geometry. This visually confirms the MAE results: the filter successfully leverages the velocity state established by the camera to smooth the noisy acoustic inputs, extending reliable tracking well beyond the physical line-of-sight.

D. Discussion

1) *Performance Trade-offs: DAS Noise vs. Camera Precision*: The results from Study A (Figure 8) provide the fun-

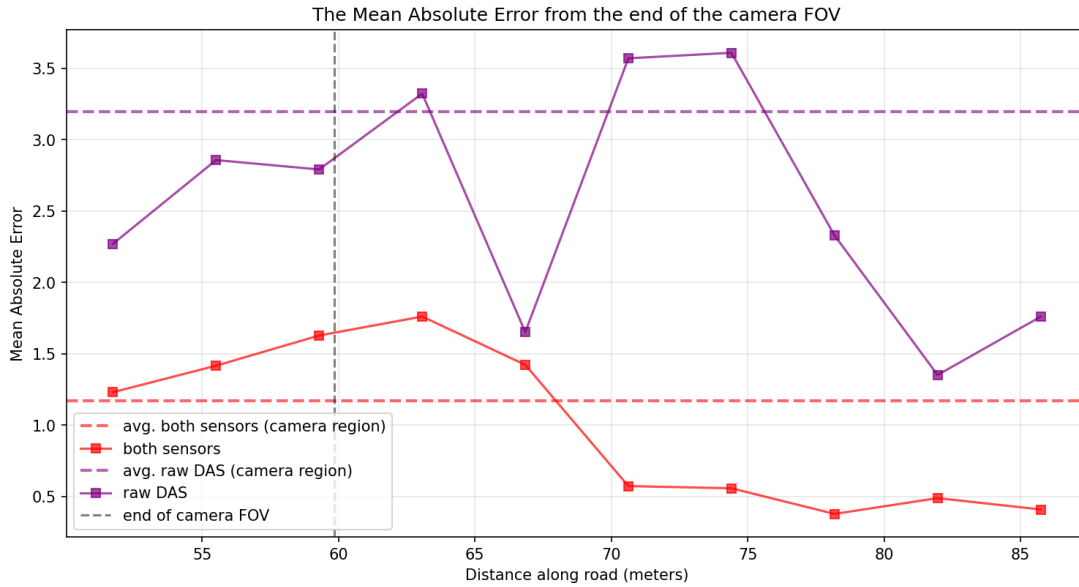


Fig. 12: Real-World Continuity Analysis.

We artificially cut camera data at $x = 55$. The Fusion model (Orange) maintains a stable, low-error track, significantly smoothing the noisy Raw DAS input (Purple).

damental justification for this sensor fusion approach. While Raw DAS exhibits an intrinsically high Mean Absolute Error ($\approx 2.4\text{m}$), the Fusion model still successfully suppresses this noise. Inside the camera’s Field of View (FOV), the high precision of the vision system allows the Kalman Filter to converge on a stable and accurate state vector. Crucially, this high-precision state estimate is then carried over into the DAS-only blind zone, where the filter acts as a sophisticated low-pass filter on the noisy DAS updates. This carry-over maintains the state estimate error low even beyond the camera FOV limit, demonstrating that the fusion system is not simply an average of the two sensors but a significant improvement over the performance of either sensor in isolation since their properties complement each other, with the high-precision camera calibrating the Kalman filter state within the FOV, enabling the continuous-coverage DAS to sustain the low-error track in the blind zone. The camera provides the precision necessary to initialize the high-quality state, and the DAS provides the continuity necessary to sustain the track.

2) *Infrastructure Optimization and the Point of Diminishing Returns*: Simulation Study B provides a clear, cost-effective infrastructure deployment guideline. In the Sparse Regime ($N = 1$ camera), the Fusion model is essential, delivering a $> 100\%$ HOTA improvement over the Vision-Only baseline by leveraging DAS to achieve a high recall ($\approx 90\%$). This directly addresses the motivation of track fragmentation in conventional systems. The analysis identified the Saturation Point at $N = 4$ cameras, corresponding to a camera-to-meter ratio of one camera per ≈ 75 meters (along a 300m corridor). At this density, the visual coverage is sufficient (88% recall), and the addition of DAS provides negligible benefit. Furthermore, the analysis of the Dense Regime ($N = 6$) highlights a critical finding: when high-precision CV data is available continuously, incorporating the inherently noisier

DAS data degrades the overall performance (HOTA drops from 83.20 to 54.95). When taking in account additional costs of increasing the camera network density, This confirms that the sparse camera (e.g., 100m spacing) with our Fusion model deployment strategy maximizes the cost-benefit ratio.

3) *Outlook and Future Work*: While the current Fusion architecture system successfully demonstrates the feasibility of high-fidelity tracking in sparse camera networks, several avenues remain for future exploration.

- **DAS Signal Processing model improvement**: In our current study, we neither used experiment data with or simulated weather conditions (fog, rain, sand storm), lighting conditions change (night). Introducing these would further the point of this work since they only take from the CV model performance.
- **The CFOS model is based on the UNet architecture of an Encoder-Decoder, but with ResNet18 as an Encoder**. These architectures are not new, specifically the UNet++ is a good Decoder, but there are better choices for Encoders currently. If we move away from Encode-Decoder Segmentation models, newer Transformer-based segmentation architectures show great results and might perform better.
- **Non-Linear Filter Optimization**: The current system utilizes a Discrete Kalman Filter, which relies on the system being Linear and the errors being Gaussian. Future work could tackle the Linear assumption and explore the use of an Extended Kalman Filter (EKF) - a state estimation algorithm for nonlinear dynamic systems. Another good option would be to tackle the Gaussian error assumption with a Particle Filter to explicitly model the non-linear transformation between pixel and world coordinates and the non-Gaussian characteristics of the YOLO output (quantization, heavy-tailed outliers), which could further

improve the state estimate's accuracy.

- Adaptive Noise Covariance: Other methods for calculating this covariance could be explored, such as incorporating temporal correlation or directly estimating the noise power from non-detection channels, or maybe non linear calculation.
- Newer YOLO model: The YOLOv8 CV model we used is relatively new and was released on January 2023, but there are already significant improvements in the newer YOLOv11 model. Also there is a variety of CV models to use, this is mainly for the training of the DAS processing model since it needs to be trained from scratch on recorded real world data.
- Lateral Tracking and Lane Changing and complex road: The current validation focuses on a simple straight road, without turns, stop lights, traffic signals, crosswalks, with a single lane for each direction, performing longitudinal tracking.
Future work could tackle each one of these to fully validate the system's precision. Particularly for complex traffic statistics like lane-changing events or vehicle interactions, which demand higher lateral precision. For this, additional data needs to be collected on various local municipality road network topologies.
- Tracking and ID consolidation algorithms: Currently we implemented a very simple tracking algorithm to maintain continuous vehicle IDs extracted from the CFOS model output, which could be improved. Furthermore, YOLO has a built in ID tracking feature, but The conversion of the YOLO detections to the 2D road coordinates we demonstrated allows for implementation of more complex continuous tracking algorithms which would certainly have better results.

VI. CONCLUSION

The Fusion architecture successfully demonstrated a novel, robust method for single-vehicle tracking by fusing the high spatial precision of Computer Vision (CV) with the continuous temporal coverage of Distributed Acoustic Sensing (DAS) within a Kalman Filter framework. Simulation confirmed that the system maintains a mean localization error below 1.5m for up to 100m beyond the camera's Field of View, establishing a theoretical efficient sparse camera density of approximately one node per 100 meters. Real-world validation verified that the fusion framework significantly improves tracking continuity compared to both Raw YOLO and Vision-Only Kalman baselines, primarily by leveraging the continuous DAS input to prevent track fragmentation and state drift. This work lays the foundation for future scalable, cost-effective urban traffic management systems that would leverage the existing optic-fiber infrastructure and the DAS system.

APPENDIX

CFOS PRE PROCESSING PIPELINE YOLO CONFIDENCE SIMULATION AND SIGMA CALCULATION

Euclidean distance (radius): $\mathbf{r} = |\mathbf{X}_{gt} - \mathbf{X}_{cam}|$

Intermediate confidence (Plateau Gaussian):

$$C_{int}(\mathbf{r}) = 100 \cdot \min \left(1, \exp \left(-\frac{1}{2} \left(\frac{\max(0, \mathbf{r} - R_0)}{k} \right)^2 \right) \right)$$

where: $k = \frac{r_{keep} - R_0}{\sqrt{-2 \ln(C_{th}/100)}}$

Final confidence with noise:

$$C_{final} = \text{clip} (C_{int}(\mathbf{r}) + \mathcal{N}(0, 3^2), 0, 100)$$

Sigma (Uncertainty) Conversion:

$$\Sigma = \text{sig_mult} \cdot \left(\frac{1}{C_{final}} \right) + \text{sig_bias}$$

CAMERA SIMULATION SPATIAL ERROR MODEL

Image projection:

$$\mathbf{u} = \Pi(K, R, t, \mathbf{X})$$

$$\mathbf{u} = \begin{bmatrix} u \\ v \end{bmatrix}, \quad u = \frac{f_x y_{cam}}{x_{cam}} + c_x, \quad v = \frac{f_y z_{cam}}{x_{cam}} + c_y.$$

Quantization to pixel centers:

$$\tilde{\mathbf{u}} = \lfloor \mathbf{u} \rfloor + \frac{1}{2}.$$

Back-projection to world (plane $z = 0$):

$$\tilde{\mathbf{X}} = \mathbf{H}^{-1} \tilde{\mathbf{u}}, \quad \mathbf{X}_{cam} = \mathbf{R}\mathbf{X} + \mathbf{t}.$$

Spatial quantization error:

$$\Delta \mathbf{X} = \tilde{\mathbf{X}} - \mathbf{X},$$

$$|\Delta \mathbf{X}| \uparrow \text{ as } x_{cam} \uparrow.$$

CAMERA SIMULATION TEMPORAL ERROR MODEL

Nominal downsampled times:

$$\mathbf{t}^{\text{nom}} = \{t_k^{\text{nom}}\}_{k=0}^{N-1}.$$

$$T_{\min} = \frac{\text{tmin_multiplier}}{\text{cam_sample_rate}}.$$

Half-normal jitter:

$$\epsilon_k \sim |\mathcal{N}(0, \sigma^2)|$$

$$\mathbf{t}_k^{(0)} = t_k^{\text{nom}} + \epsilon_k.$$

Monotonicity and minimum gap:

$$\mathbf{t}_0^{(1)} = \mathbf{t}_0^{(0)}$$

$$\mathbf{t}_k^{(1)} = \max(\mathbf{t}_{k-1}^{(1)} + T_{\min}, t_k^{\text{nom}}, \mathbf{t}_k^{(0)}).$$

Span-preserving rescale of excess lag:

$$s = \frac{t_{N-1}^{\text{nom}} - t_0^{\text{nom}}}{\mathbf{t}_{N-1}^{(1)} - \mathbf{t}_0^{(1)}} \\ \mathbf{t}_k^{(2)} = t_k^{\text{nom}} + s(\mathbf{t}_k^{(1)} - t_k^{\text{nom}}).$$

No-early final timestamps:

$$\mathbf{t}_k = \max(\mathbf{t}_k^{(2)}, t_k^{\text{nom}}) \\ \text{so } \mathbf{t}_k \geq t_k^{\text{nom}}, \quad \mathbf{t}_k - \mathbf{t}_{k-1} \geq T_{\min}.$$

DKF FUNDAMENTALS

The system dynamics and measurements are modeled as:

$$\mathbf{x}_k = \mathbf{F}_k \mathbf{x}_{k-1} + \mathbf{w}_k \quad (\text{State Equation}) \quad (7)$$

$$\mathbf{z}_k = \mathbf{H}_k \mathbf{x}_k + \mathbf{v}_k \quad (\text{Measurement Equation}) \quad (8)$$

Where $\mathbf{x}_k$ is the state vector and $\mathbf{z}_k$ is the measurement vector. The process noise $\mathbf{w}_k$ and measurement noise $\mathbf{v}_k$ are assumed to be zero-mean Gaussian white noise with known covariance matrices $\mathbf{Q}_k$ and $\mathbf{R}_k$, respectively.

The prediction step projects the state estimate $\hat{\mathbf{x}}_{k-1|k-1}$ and error covariance $\mathbf{P}_{k-1|k-1}$ forward:

$$\hat{\mathbf{x}}_{k|k-1} = \mathbf{F}_k \hat{\mathbf{x}}_{k-1|k-1} \quad (9)$$

$$\mathbf{P}_{k|k-1} = \mathbf{F}_k \mathbf{P}_{k-1|k-1} \mathbf{F}_k^T + \mathbf{Q}_k \quad (10)$$

The measurement update incorporates new data $\mathbf{z}_k$ via the Kalman Gain $\mathbf{K}_k$:

$$\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}_k^T (\mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^T + \mathbf{R}_k)^{-1} \quad (11)$$

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k (\mathbf{z}_k - \mathbf{H}_k \hat{\mathbf{x}}_{k|k-1}) \quad (12)$$

$$\mathbf{P}_{k|k} = (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k) \mathbf{P}_{k|k-1} \quad (13)$$

YOLO STATIONARY OBJECT REMOVAL ALGORITHM

Let a track consist of a sequence of detections, (t_i, x_i, y_i) , where $i \in \{1, \dots, N\}$. We define a window size W (e.g., $W = 4$ frames) for averaging.

- The ****Start Position**** $(\bar{x}_{\text{start}}, \bar{y}_{\text{start}})$ is the average of the first W detection coordinates:

$$\bar{x}_{\text{start}} = \frac{1}{W} \sum_{i=1}^W x_i, \quad \bar{y}_{\text{start}} = \frac{1}{W} \sum_{i=1}^W y_i$$

- The ****End Position**** $(\bar{x}_{\text{end}}, \bar{y}_{\text{end}})$ is the average of the last W detection coordinates:

$$\bar{x}_{\text{end}} = \frac{1}{W} \sum_{i=N-W+1}^N x_i, \quad \bar{y}_{\text{end}} = \frac{1}{W} \sum_{i=N-W+1}^N y_i$$

The total Average Pixel Displacement Δu is then calculated using the simple Euclidean distance between the start and end positions:

$$\Delta u = \sqrt{(\bar{x}_{\text{end}} - \bar{x}_{\text{start}})^2 + (\bar{y}_{\text{end}} - \bar{y}_{\text{start}})^2}$$

A track is considered moving and is retained only if the calculated displacement Δu is greater than or equal to a minimum displacement threshold $U_{\min}$ (e.g., $U_{\min} = 40$

pixels). This serves as the sole filtering condition for stationary objects:

$$\Delta u \geq U_{\min}$$

Objects failing this condition are rejected as likely stationary detections that were merely jittering in place. The parameters used in this process are window size W and the minimum displacement $U_{\min}$.

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(a) Occlusion Resilience.

Comparison of Raw YOLO (Blue/Y-2), Vision-Only KF (Yellow/KY-2), and Fusion (Green/KB-2). The Fusion model maintains the most accurate bounding box during visual clutter.



(b) Geospatial Verification.

Green dots represent the Fusion Kalman Filter state estimates on Klauzner St., Tel Aviv. The track remains smooth and lane-aligned even beyond the effective camera range ($> 90\text{m}$).

Fig. 13: Qualitative Evaluation.

- (a) In complex visual environments, the Fusion model resists “distractions” from static objects better than vision-only baselines. (b) On the street map, the fused trajectory demonstrates smooth continuity extending into the non-visual zone.

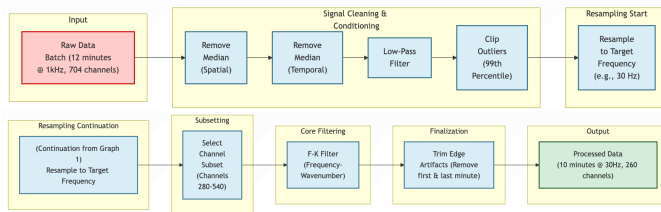


Fig. 14: DAS Pre Processing procedure flow diagram.

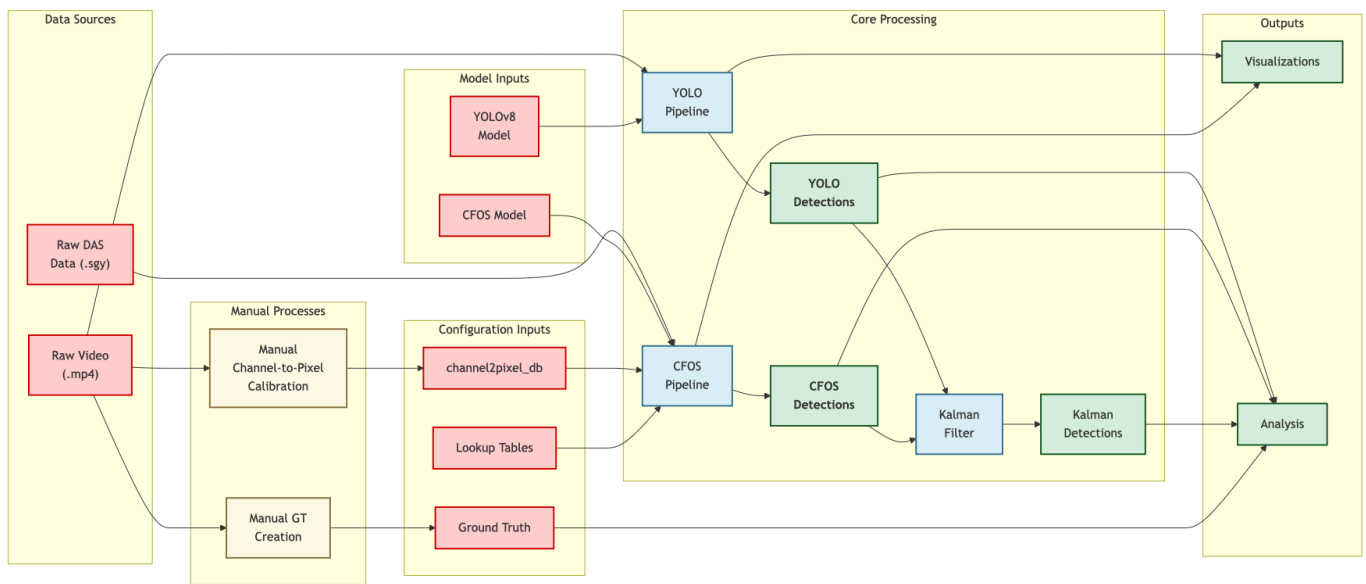


Fig. 15: Architecture Signal Processing Flow Diagram